

# Appendix for Optimised fitness functions for automated runtime improvement of software

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## Appendix A

A visual representation of the variants examined and their fitness values in 2 different scenarios. One with the task\_clock measure and a retry setting of 1 and the other with the weights measure and a retry of 2. We can observe how different measures and retry setting affect the exploration of the search space.

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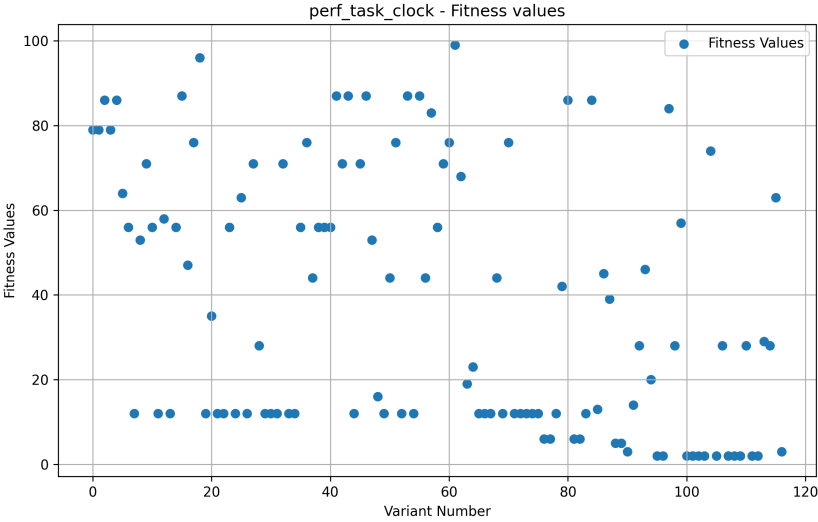
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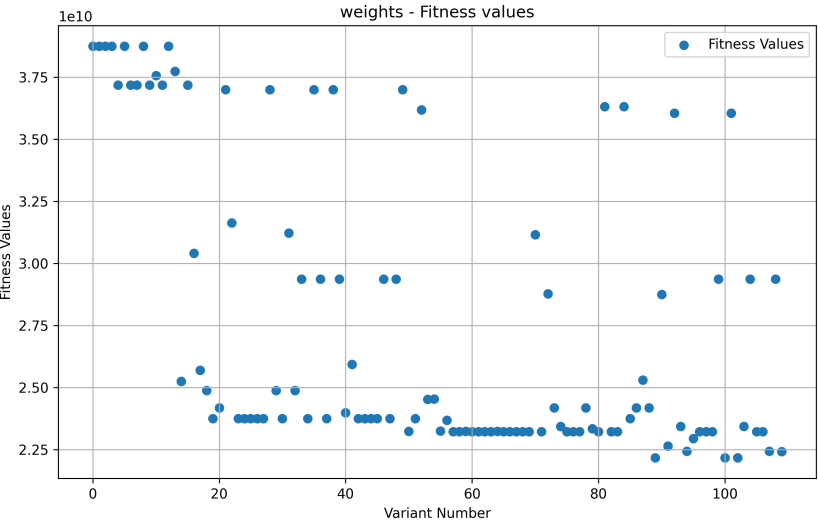
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Fig. 1. Comparative analysis of the search space for the LPG benchmark under different measures and retries.



(a) Search space exploration with task\_clock and retry=1



(b) Search space exploration with 'weights' measure and retry=2

Appendix B

In this section we present the results of running the same exact experiments as seen in Section 5 with the only difference being that MAGPIE was invoked with the Genetic Programming approach as the search strategy.

Table 1. RQ2: Mean Median Execution Times, Median Ranks, and Number of Variants for Each Measure

| Measure          | Mean MET(s) | Median Rank | NoV |
|------------------|-------------|-------------|-----|
| Original         | 9.65        | 13          | -   |
| Energy           | 6.21        | 11          | 257 |
| L1_dcache_loads  | 5.99        | 7           | 397 |
| branch_misses    | 5.75        | 5           | 431 |
| branches         | 6.22        | 7           | 393 |
| cache_misses     | 7.02        | 8           | 419 |
| cache_references | 4.64        | 5           | 451 |
| cpu_clock        | 8.85        | 14          | 376 |
| cycles           | 6.15        | 4           | 426 |
| faults           | 7.28        | 12          | 381 |
| instructions     | 6.48        | 7           | 384 |
| task_clock       | 9.87        | 14          | 387 |
| perf_time        | 5.66        | 7           | 409 |
| Time             | 5.17        | 4           | 414 |
| Weights          | 6.96        | 12          | 275 |

**MET:** Median Execution Time  
**NoV:** Number of Variants examined

Table 2. RQ3: Statistical Analysis of Retry Rankings Across All Benchmarks

| Retries | Median Rank | Mean Rank | STD  | IQR  |
|---------|-------------|-----------|------|------|
| 1       | 2.36        | 2.47      | 0.27 | 0.39 |
| 2       | 2.93        | 2.95      | 0.40 | 0.50 |
| 3       | 3.00        | 3.07      | 0.46 | 0.71 |
| 4       | 3.29        | 3.18      | 0.41 | 0.39 |
| 5       | 3.36        | 3.33      | 0.41 | 0.32 |

Table 3. Recommended Retry Numbers for Each Measure

| Measure          | Mean RN | Median RN | IQR | STD  |
|------------------|---------|-----------|-----|------|
| Energy           | 3.0     | 3.0       | 1.0 | 1.10 |
| L1_dcache_loads  | 2.64    | 2.0       | 2.5 | 1.57 |
| branch_misses    | 2.64    | 2.0       | 2.0 | 1.50 |
| branches         | 2.18    | 2.0       | 1.5 | 1.54 |
| cache_misses     | 3.0     | 3.0       | 4.0 | 1.79 |
| cache_references | 2.18    | 2.0       | 1.5 | 1.33 |
| cpu_clock        | 3.45    | 3.0       | 2.5 | 1.44 |
| cycles           | 2.45    | 3.0       | 2.0 | 1.37 |
| faults           | 2.09    | 2.0       | 1.5 | 1.22 |
| instructions     | 2.55    | 2.0       | 3.0 | 1.51 |
| task_clock       | 3.64    | 4.0       | 1.5 | 1.29 |
| perf_time        | 2.82    | 3.0       | 2.5 | 1.54 |
| Time             | 2.73    | 2.0       | 3.5 | 1.74 |
| Weights          | 2.27    | 2.0       | 2.0 | 1.35 |

RN: Retry Number

Table 4. Percentage Decrease in Variants Explored by Number of Retries

| Retries | Mean Decr.% | Median Decr. % | STD   | IQR   |
|---------|-------------|----------------|-------|-------|
| 1       | 0.0         | 0.0            | 0.0   | 0.0   |
| 2       | 31.87       | 34.84          | 24.64 | 38.11 |
| 3       | 52.86       | 60.07          | 22.30 | 25.65 |
| 4       | 64.73       | 69.93          | 19.58 | 21.00 |
| 5       | 72.40       | 75.58          | 15.14 | 15.63 |

Table 5. Median and Mean MAD for Each Measure

| Measure         | Med. MAD | Mean MAD | STD   | IQR   |
|-----------------|----------|----------|-------|-------|
| Energy          | 0.085    | 0.097    | 0.230 | 0.234 |
| L1_dcache_loads | 0.140    | 0.129    | 0.224 | 0.229 |
| branch_misses   | 0.126    | 0.121    | 0.190 | 0.176 |
| branches        | 0.183    | 0.162    | 0.246 | 0.245 |
| cache_misses    | 0.087    | 0.077    | 0.159 | 0.152 |
| cache_refs      | 0.186    | 0.175    | 0.251 | 0.280 |
| cpu_clock       | 0.461    | 0.533    | 0.150 | 0.123 |
| cycles          | 0.170    | 0.145    | 0.210 | 0.191 |
| faults          | 0.022    | 0.025    | 0.036 | 0.040 |
| instructions    | 0.142    | 0.139    | 0.260 | 0.257 |
| task_clock      | 0.428    | 0.498    | 0.165 | 0.131 |
| perf_time       | 0.122    | 0.150    | 0.248 | 0.224 |
| Time            | 0.210    | 0.169    | 0.252 | 0.257 |
| Weights         | 0.153    | 0.159    | 0.340 | 0.350 |

Table 6. Median and Mean MAD for Each Retry Number

| Retries | Med. MAD | Mean MAD | STD   | IQR   |
|---------|----------|----------|-------|-------|
| 1       | 0.261    | 0.311    | 0.206 | 0.249 |
| 2       | 0.197    | 0.242    | 0.189 | 0.192 |
| 3       | 0.176    | 0.223    | 0.194 | 0.207 |
| 4       | 0.146    | 0.188    | 0.168 | 0.207 |
| 5       | 0.141    | 0.193    | 0.169 | 0.242 |

Table 7. Analysis of Retry Parameters on Search Efficiency

| Retries | TD  | ADPS   | POD    |
|---------|-----|--------|--------|
| 1       | 8.0 | 0.0500 | 0.0123 |
| 2       | 7.0 | 0.0563 | 0.0178 |
| 3       | 6.0 | 0.0521 | 0.0251 |
| 4       | 6.0 | 0.0497 | 0.0303 |
| 5       | 5.5 | 0.0642 | 0.0365 |

**TD:** Total decreases  
**ADPS:** Average Decrease Percentage per Step  
**POD:** Proportion of Decreases

Table 8. Performance of Each Measure in the Search Process

| Measure          | TD  | ADPS   | POD    |
|------------------|-----|--------|--------|
| Energy           | 6.0 | 0.0291 | 0.0383 |
| L1_dcache_loads  | 8.0 | 0.0343 | 0.0253 |
| branch_misses    | 8.0 | 0.0486 | 0.0215 |
| branches         | 8.0 | 0.0429 | 0.0254 |
| cache_misses     | 6.0 | 0.0324 | 0.0232 |
| cache_references | 7.0 | 0.0571 | 0.0209 |
| cpu_clock        | 4.0 | 0.2337 | 0.0149 |
| cycles           | 8.0 | 0.0497 | 0.0231 |
| faults           | 7.0 | 0.0129 | 0.0255 |
| instructions     | 7.0 | 0.0384 | 0.0296 |
| task_clock       | 5.0 | 0.1791 | 0.0153 |
| perf_time        | 7.0 | 0.0439 | 0.0254 |
| Time             | 7.0 | 0.0564 | 0.0229 |
| Weights          | 5.0 | 0.0792 | 0.0329 |

**TD:** Total decreases  
**ADPS:** Average Decrease Percentage per Step  
**POD:** Proportion of Decreases